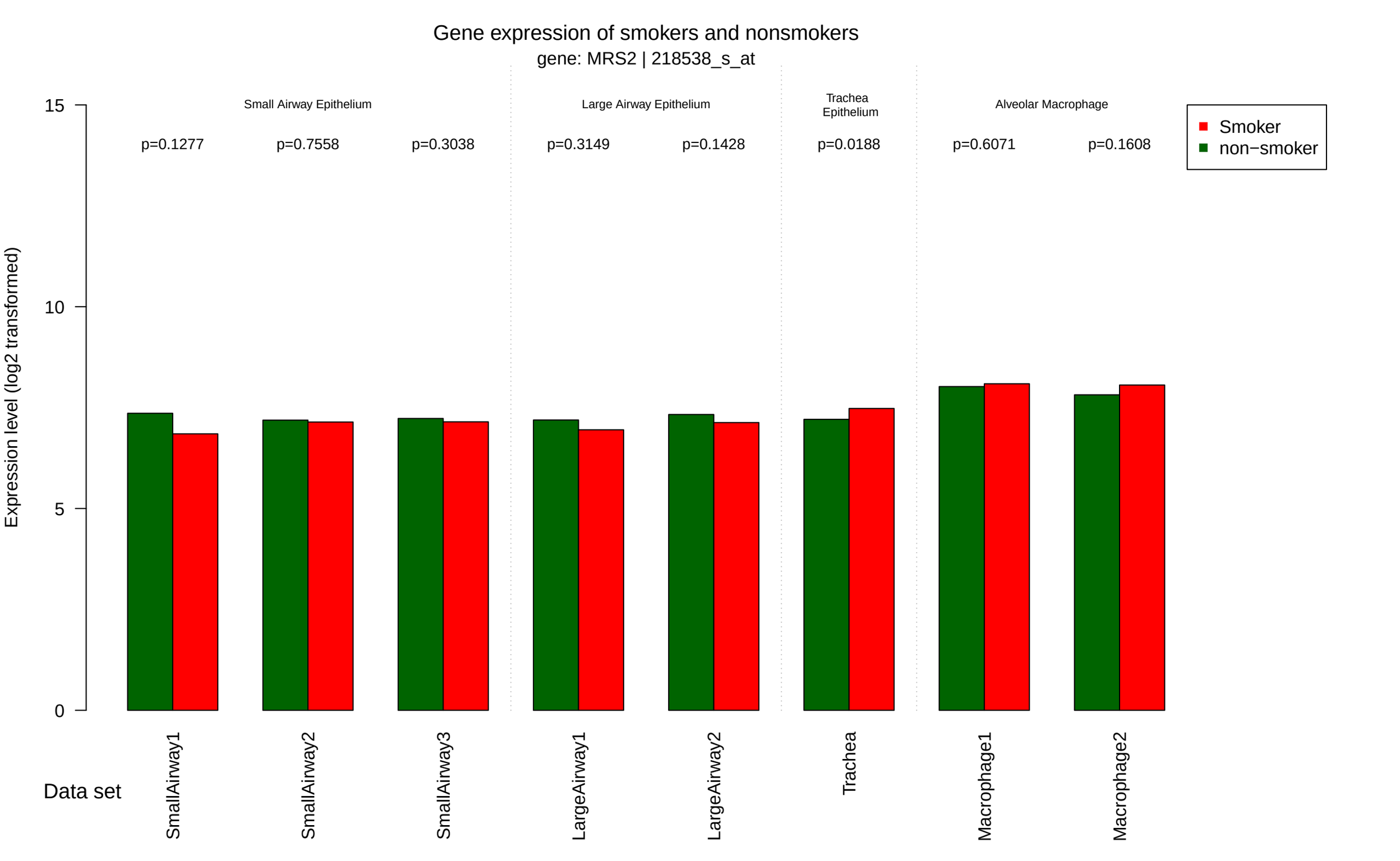
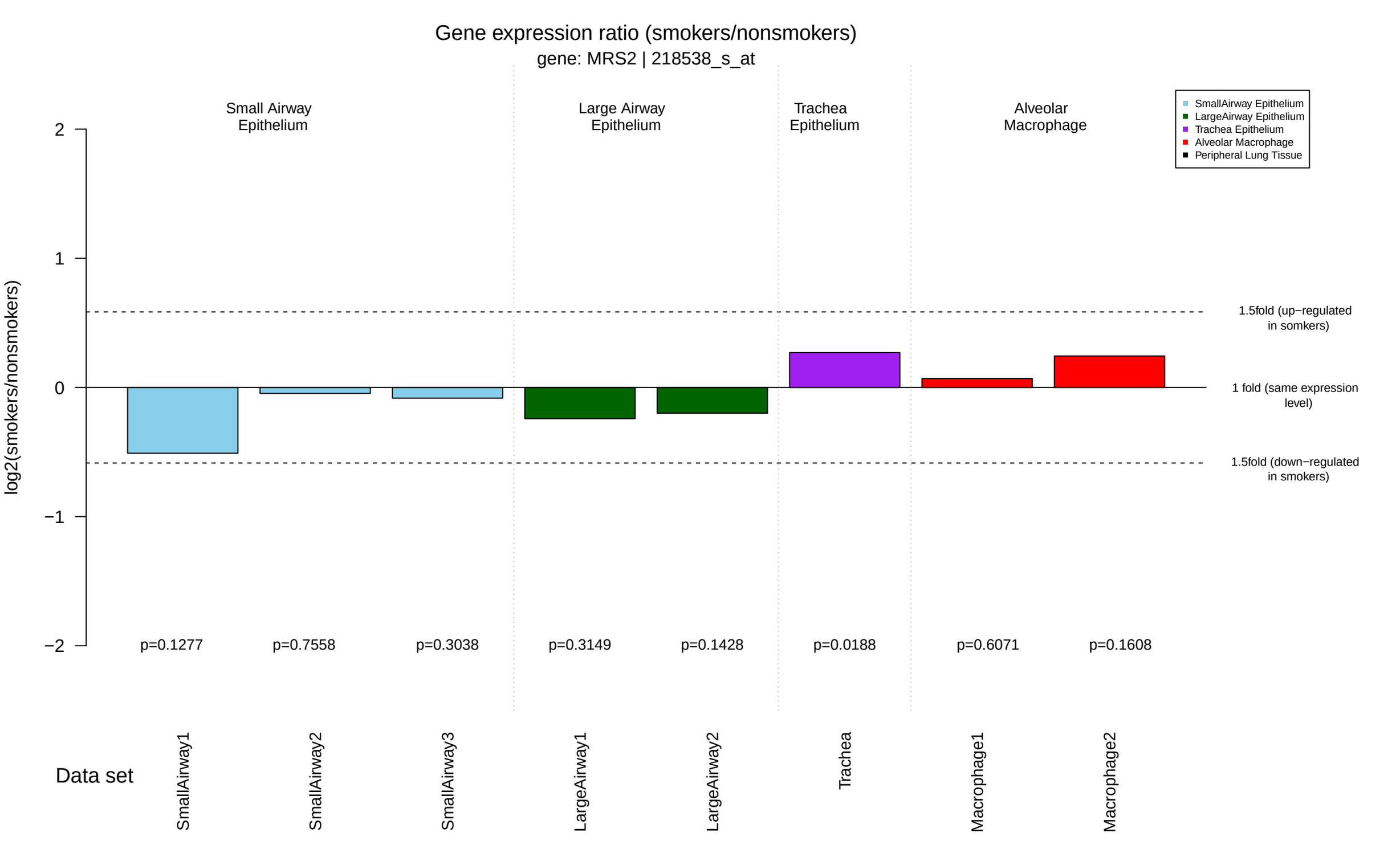
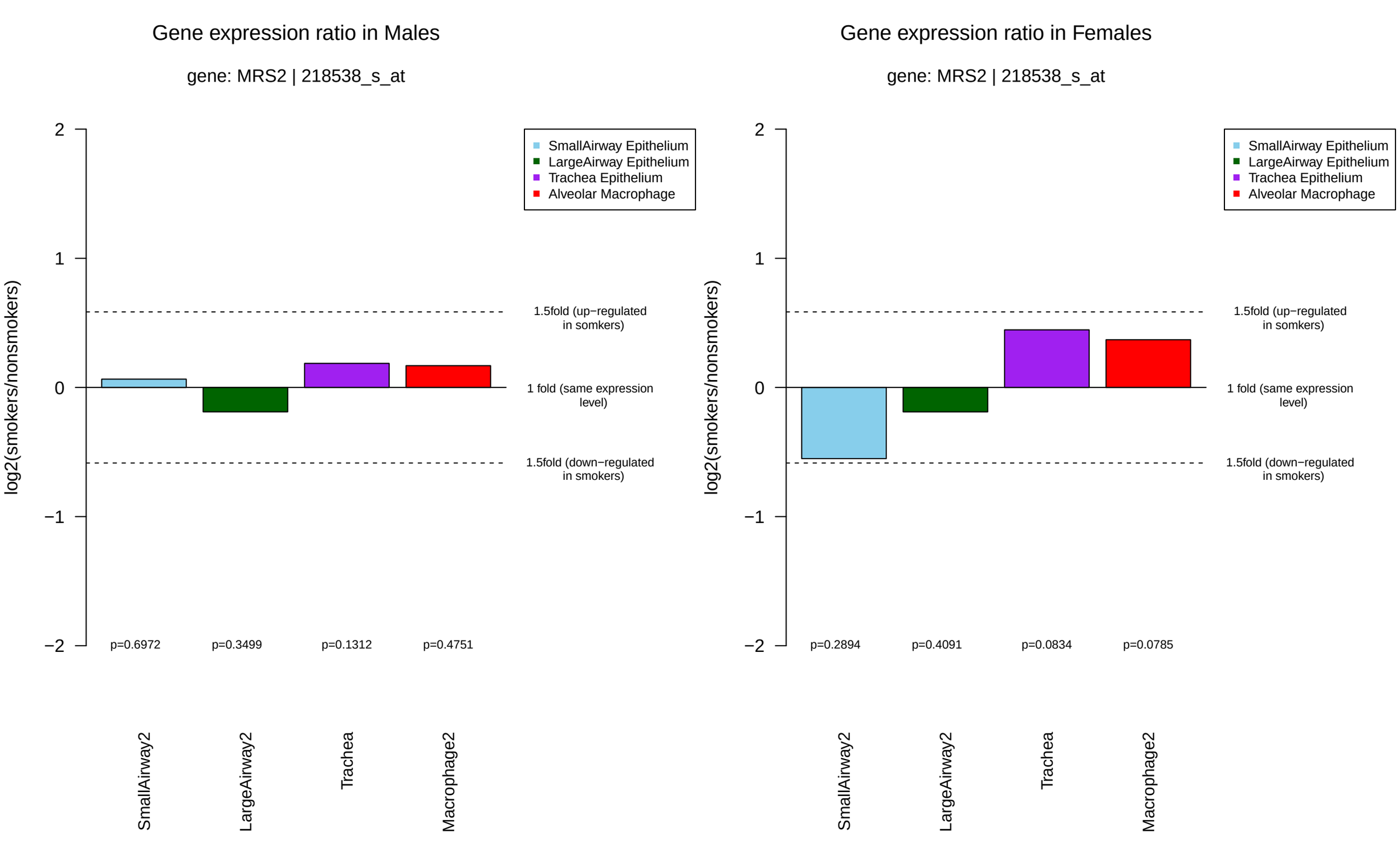
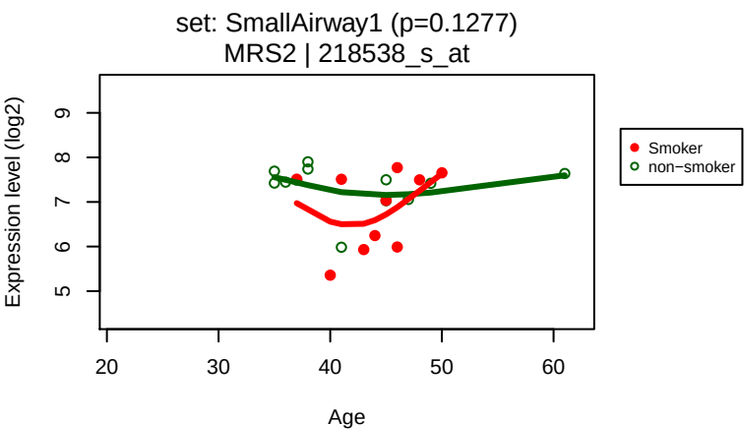
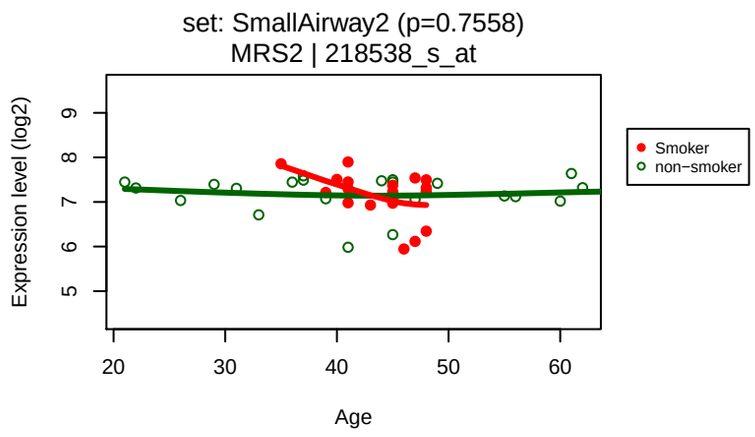
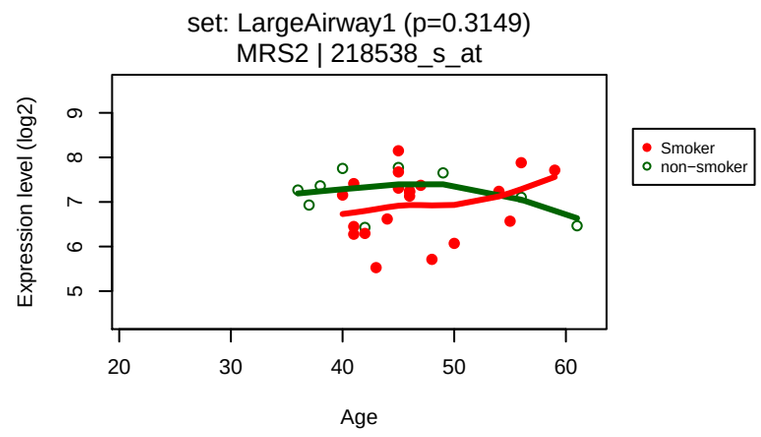

Gene expression of smokers and nonsmokers
gene: MRS2 | 218538_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1277
p=0.7558
p=0.3038
p=0.3149
p=0.1428
p=0.0188
p=0.6071
p=0.1608
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

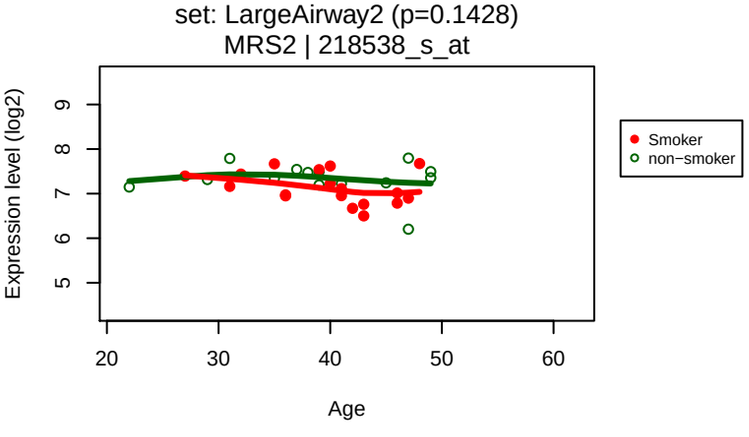

Gene expression ratio (smokers/nonsmokers)
gene: MRS2 | 218538_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1277
p=0.7558
p=0.3038
p=0.3149
p=0.1428
p=0.0188
p=0.6071
p=0.1608
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

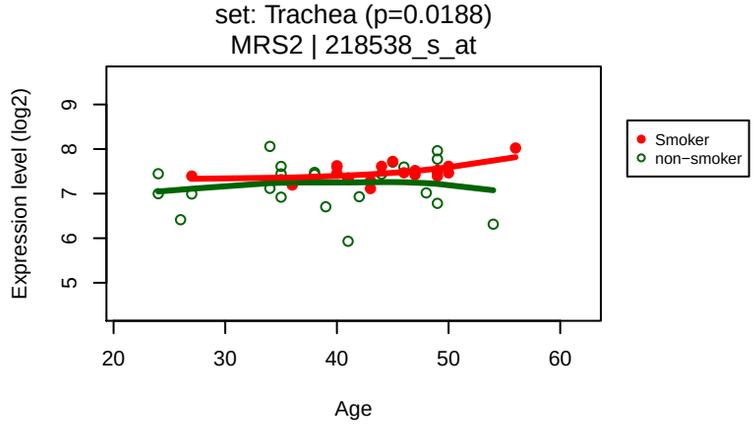

Gene expression ratio in Males
gene: MRS2 | 218538_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6972
p=0.3499
p=0.1312
p=0.4751
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRS2 | 218538_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2894
p=0.4091
p=0.0834
p=0.0785
SmallAirway2
LargeAirway2
Trachea
Macrophage2

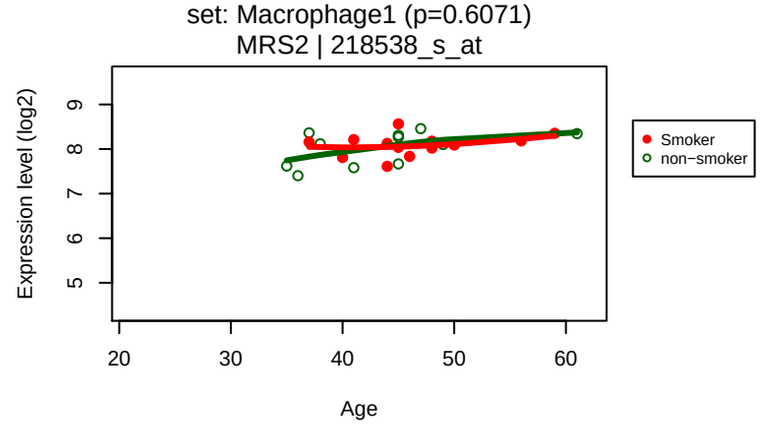

set: SmallAirway1 (p=0.1277)
MRS2 | 218538_s_at
Expression level (log2)
Age
Smoker
non−smoker

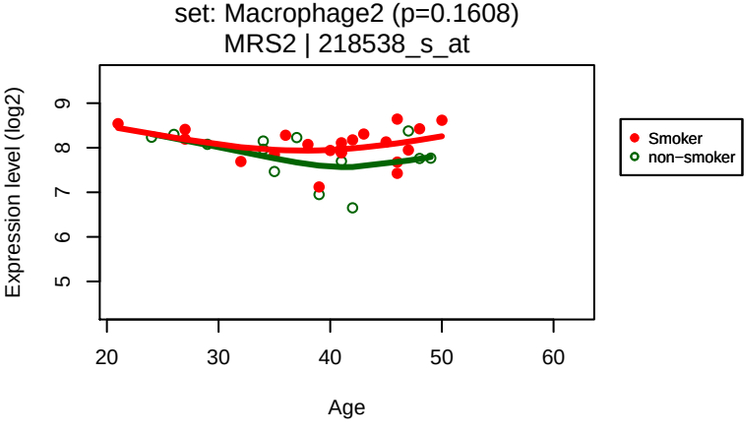

set: SmallAirway2 (p=0.7558)
MRS2 | 218538_s_at
Expression level (log2)
Age
Smoker
non−smoker

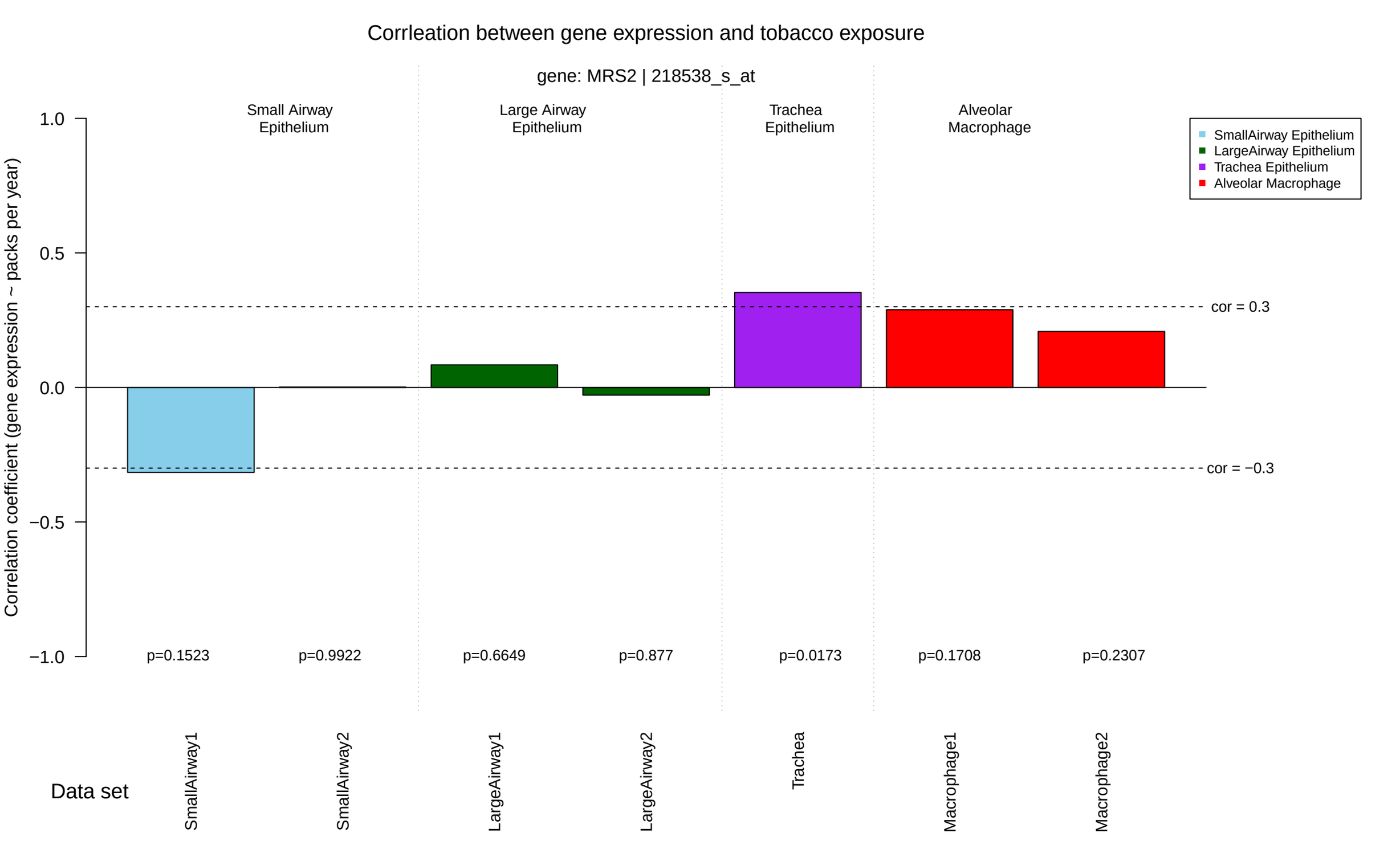

set: LargeAirway1 (p=0.3149)
MRS2 | 218538_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1428)
MRS2 | 218538_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0188)
MRS2 | 218538_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6071)
MRS2 | 218538_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1608)
MRS2 | 218538_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRS2 | 218538_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1523
p=0.9922
p=0.6649
p=0.877
p=0.0173
p=0.1708
p=0.2307
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2